Thesis for Master's Degree

# State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

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2025

석 사 학 위 논 문

# 라그랑지안 기반 제약 강화학습을 이용한 자율주행에서의 상태별 안전성 고려

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2025

# State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

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Gwangju, Republic of Korea

June 20, 2025

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# State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

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the degree of Master of Science

June 20, 2025

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Dedicated to my family.

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## Abstract

For the practical deployment of autonomous driving systems, high levels of safety and adaptability are essential. Deep Reinforcement Learning (DRL), which allows an agent to learn driving strategies through interaction with the environment, has attracted significant attention. However, DRL can raise safety concerns when applied in real-world scenarios due to challenges such as reward design and poor generalization to out-of-distribution situations. To address these issues, Constrained Reinforcement Learning (CRL) has been proposed, which seeks to balance safety and performance during policy learning. CRL is typically formulated using Constrained Markov Decision Processes (CMDPs), where constraints are defined as expected cumulative costs. Nevertheless, in autonomous driving, some constraints—such as collision avoidance—require immediate reactions. Thus, even when CMDP-based policies satisfy cumulative cost-based constraints, violations of instant constraints may still occur. In this paper, we propose a Lagrangian-based policy optimization method that estimates state-wise Lagrangian multipliers to precisely reflect constraints and consider state-level safety. The

proposed method is validated on a navigation task in the Safety Gym environment developed by OpenAI, where a vehicle must reach a designated goal location.

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# 국 문 요 약

자율주행 시스템의 실제 적용을 위해서는 높은 안정성과 적응성이 요구된다. 이를 위해 환경과의 상호작용을 통해 스스로 주행 전략을 학습하는 심층 강화 학습(Deep Reinforcement Learning, DRL)이 주목받고 있다. 그러나 심층 강화 학습은 보상 설계, 학습 분포 외의 상황에 대한 일반화 실패 등으로 인해 실제 적용 시 안전 문제가 발생할 수 있다. 이를 해결하고자 정책 학습 시 안정성과 성능 간의 균형을 도모하는 제약 강화 학습(Constrained Reinforcement Learning, CRL)이 제안되었다. 제약 강화 학습은 일반적으로 기댓값 기반 누적 비용 형태로 제약 조건을 정의하는 제약 마르코프 결정(Constrained Markov Decision Process, CMDP)에 기반하는데 자율주행에서는 충돌 회피와 같이 즉각적인 반응이 요구되는 제약 조건들이 존재하기 때문에, CMDP에서 정의한 누적 제약을 만족하도록 학습하더라도 이러한 즉각적인 제약을 위반하는 상황이 발생할 수 있다. 따라서 본 논문에서는 라그랑지안 기반 정책 최적화 방법을 통해, 상태별로 라그랑주 승수(Lagrange multiplier)를 추정하여 제약 조건을 정밀하게 반영함으로써 상태별 안정성을 고려하고자 한다. 제안한 방법은 OpenAI의 시뮬레이션 환경인 Safety Gym에서 차량이 지정된 목표 지점까지 이동하는 태스크를 통해 검증하였다.

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# Chapter 1

# Introduction

## 1.1 Introduction

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## 1.2 Motivation

insert your own materials [1].

| | Overall classification results (%) | | |
|---|---|---|---|
| Descriptors | KTH | UCF sports | TV human interaction |
| Trajectory | 91.67 | 74.64 | 46.43 |
| HOG | 86.11 | 69.88 | 44.52 |
| MBH | 96.3 | 72.74 | 45.88 |
| Combined | 96.3 | 81.07 | 55.4 |

Table 1.1: An example table

Figure 1.1: The symbol of GIST.

# Chapter 2

# The chapter 2

Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2. Write your chapter 2.

## 2.1 A section

### 2.1.1 A subsection

write what you did.

Figure 2.1: An eps image

**Algorithm 1:** How to write algorithms 1

```
Data: this text
Result: how to write algorithm with LaTeX2e
initialization;
while not at end of this document do
    read current;
    if understand then
        go to next section;
        current section becomes this one;
    else
        go back to the beginning of current section;
```

Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you. Hi, hello, nice to meet you.

# Summary

State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

Summarize your thesis.

# References

1. A. Einstein, "Zur Elektrodynamik bewegter Körper. (German) [On the electrodynamics of moving bodies]," *Annalen der Physik*, vol. 322, no. 10, pp. 891–921, 1905.

# Appendix A

# Abbreviations

**GIST** Gwangju Institute of Science and Technology

**EECS** Electrical Engineering and Computer Science

**YOLO** You Only Live Once

# Appendix B

# More appendix

You can add more appendices.

# Acknowledgements

special thanks to...

모두들 감사합니다.